

UNIT – I

1. Introduction and Problem Solving

1.1 Definitions of AI

Artificial Intelligence (AI) is defined in various ways based on *thinking* and *acting*, *human-like* vs *rational*.

(1) Acting Humanly (Turing Test Approach)

AI is a machine's ability to behave like a human—communicate, reason, learn, use knowledge.

(2) Thinking Humanly (Cognitive Modeling Approach)

AI tries to model how humans think (based on psychology, neuroscience, introspection).

(3) Thinking Rationally (Laws of Thought Approach)

AI uses logic, rules, and reasoning to make correct decisions.

(4) Acting Rationally (Rational Agent Approach – Modern Definition)

AI acts to achieve the best possible outcome or maximize goal satisfaction.

Modern and most accepted definition:

👉 *AI is the study of agents that observe the environment and take actions to achieve goals rationally.*

1.2 Introduction to AI Applications

AI is used in multiple real-world fields:

- **Healthcare:** diagnosis systems, image interpretation, robotic surgery.
- **Finance:** fraud detection, algorithmic trading.
- **Education:** adaptive learning, AI tutors.
- **Robotics:** autonomous vehicles, drones, industrial robots.
- **Natural Language Processing:** chatbots, translation, speech recognition.
- **Game Development:** intelligent opponents, decision-making.
- **Expert Systems:** medical diagnosis, troubleshooting systems.
- **Computer Vision:** face recognition, object detection.

1.3 AI Techniques

AI techniques help computers solve complex tasks effectively. Key techniques:

(1) Search Techniques

- BFS, DFS, A* search
- Used in problem solving, path finding

(2) Knowledge Representation

- Semantic networks
- Frames
- Logic (propositional & predicate)
- Ontologies

(3) Reasoning Techniques

- Deductive, Inductive, Abductive
- Statistical reasoning

(4) Learning Techniques

- Supervised learning
- Unsupervised learning
- Reinforcement learning

(5) Planning

- Creating action sequences to achieve goals.
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2. Production Systems

A **production system** is a knowledge-based model used to solve problems using **rules**.

It has **three main components**:

(1) A Set of Production Rules

Each rule is of the form:

IF <condition> THEN <action>

Example:

IF temperature > 100 THEN turn_on_cooler

(2) Working Memory

Contains current facts/state of the problem.

(3) Control System

Decides the order in which rules will be applied.

Advantages:

- Simple to implement
- Highly modular
- Easy to modify rules

Applications:

- Expert systems
 - Diagnosis systems
 - Knowledge-based problem solving
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3. Control Strategies

Control strategies define **which rule** to apply **when multiple rules become eligible**.

Two main types:

(1) Forward Control (Data-Driven)

- Start from known facts
- Apply rules to reach goal
- Useful when **data is available**, but goal is not clear
- Used in: pattern recognition, sensor-based systems

(2) Backward Control (Goal-Driven)

- Start from goal
- Work backwards to find facts needed

- Useful when **goal is fixed**, but data is huge
 - Used in: diagnosis, query systems
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4. Reasoning – Forward and Backward Chaining

4.1 Forward Chaining (Inference from facts → conclusion)

- Starts with data
- Applies rules
- Keeps generating new facts
- Stops when goal is reached

Used in:

- Expert systems
 - Monitoring systems
 - Real-time decision making
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4.2 Backward Chaining (Inference from goal → facts)

- Start from goal
- Check which rule supports that goal
- Then find sub-goals
- Continue until facts in memory are matched

Used in:

- Diagnosis systems
 - Medical expert systems
 - Troubleshooting problems
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5. Intelligent Agents

5.1 Definition of Agent

An **agent** is anything that:

- **Perceives** its environment (via sensors)
- **Acts** on the environment (via actuators)

Examples:

- Robot, chatbot, thermostat, self-driving car.
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5.2 Definition of Rational Agent

A **rational agent** chooses an action that **maximizes performance** based on:

- Perception
- Built-in knowledge
- Possible actions
- Expected outcomes

👉 *A rational agent always tries to do the right thing.*

6. Types of Intelligent Agents

6.1 Simple Reflex Agents

- Act only on **current perception**
- No memory
- Use **condition-action rules**

Example:

IF obstacle_detected THEN turn_left

Limitation:

Fails in partial or complex environments.

6.2 Model-Based Reflex Agents

- Maintain **internal state/memory**
- Can handle **partial observability**

- Combine current input + stored info to choose actions

Example: robot remembering map of the room.

6.3 Goal-Based Agents

- Behave to **achieve specific goals**
- More flexible than reflex agents
- Use search and planning techniques

Example:

Self-driving car planning a route.

6.4 Utility-Based Agents

- Not only achieve goals
- But choose the **best among many goal-achieving actions**
- Use **utility function** (happiness, cost, time, risk)

Example:

Choosing the safest + fastest + cheapest route.

7. Agent Environments

Agents operate in various environments, defined by PEAS.

7.1 PEAS Description

- **Performance measure**
- **Environment**
- **Actuators**
- **Sensors**

Environment Types:

1. **Fully Observable vs Partially Observable**
2. **Deterministic vs Stochastic**
3. **Episodic vs Sequential**

4. **Static vs Dynamic**
 5. **Discrete vs Continuous**
 6. **Single agent vs Multi-agent**
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Complete Summary (Quick Revision)

Topic	Key Idea
AI Definitions	Human-like vs Rational thinking/acting
AI Applications	Healthcare, robotics, NLP, finance
AI Techniques	Search, reasoning, learning, planning
Production System	Rules + working memory + control
Control Strategies	Rule selection mechanism
Forward Chaining	Facts → Goal
Backward Chaining	Goal → Facts
Intelligent Agent	Perceives + Acts
Rational Agent	Maximizes performance
Agent Types	Reflex, Model-based, Goal-based, Utility-based
Environment Types	Observable, deterministic, static, etc.